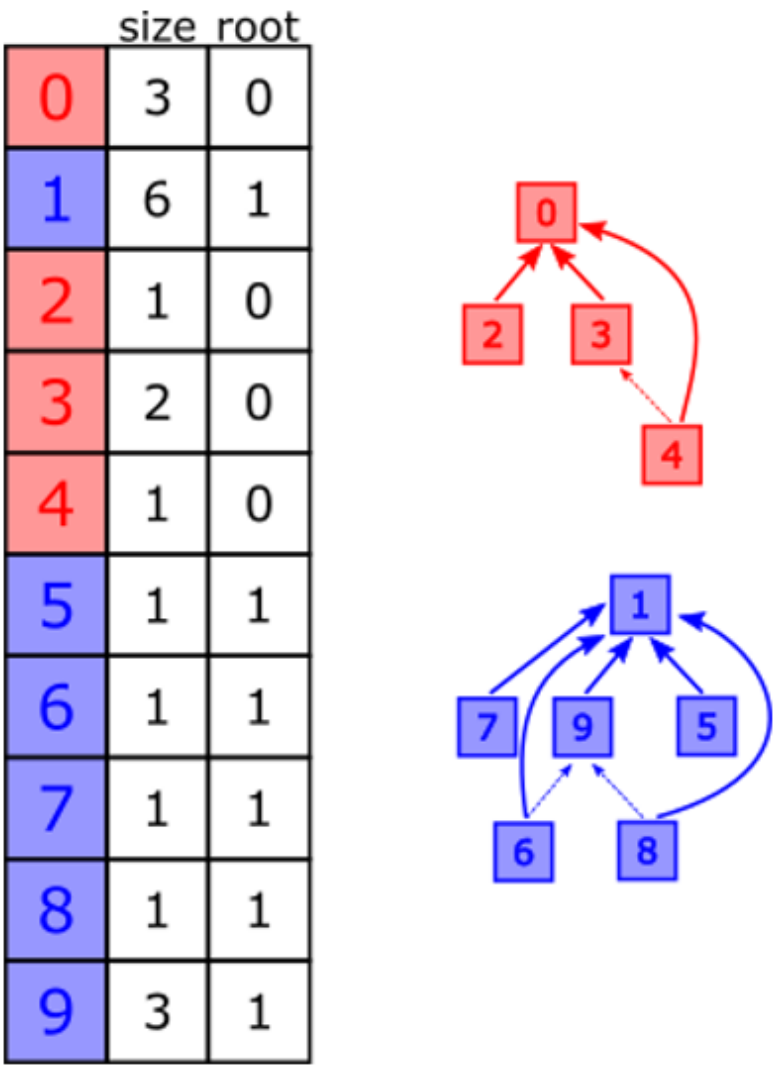


Figure 5.11 Disjoint set represented using a tree with path compression. Internal representation is shown next to the elements' array. In tree representation, dashed arrows are parent links, while solid arrows are pointers to the set’s root. The structure initially holds two sets, colored in red and blue and whose roots are respectively 0 and 1.

Now, if you were to update all the root pointers as part of merge, it wouldn't be a logarithmic method anymore: we would need linear time to update each node in the tree.